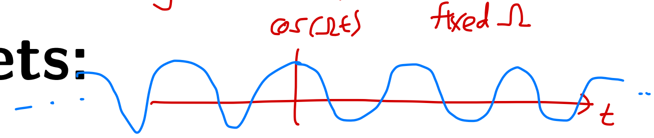
$$X(j\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt = e^{j\Omega t} \cdot x(t)$$

$$e^{j\Omega t} = \cos(\Omega t) + j \sin(\Omega t)$$

Time-Frequency and Wavelets:



- Until now, we focused on Fourier transforms (i.e. FS, CTFT, DTFT, DFT), which construct and deconstruct signals using sinusoids.
- But sinusoidal analysis has some serious limitations, as we will soon discuss.
- In this note packet, we will cover:
  - limitations of Fourier analysis
  - the short-time CTFT
  - the time/frequency uncertainty principle
  - the short-time DTFT
  - the short-time DFT
  - the continuous wavelet transform (CWT)
  - the discrete wavelet transform (DWT)
  - some applications of the DWT.

Limitations of Fourier Analysis:

- Consider the world of continuous-time signals $\{x(t)\}_{t\in\mathbb{R}}$
- For these signals, we often use the CTFT:

$$X_{\mathsf{c}}(j\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t}\,\mathrm{d}t, \quad \Omega \in \mathbb{R}$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X_{\mathsf{c}}(j\Omega)e^{j\Omega t}\,\mathrm{d}\Omega, \quad t \in \mathbb{R}.$$

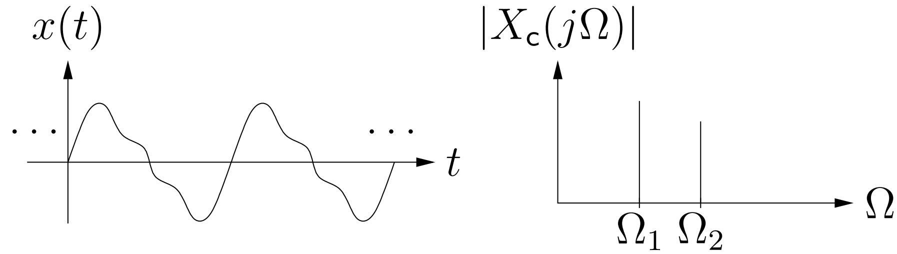
- The CTFT constructs $x(t)$ from basis elements $b_\Omega(t) \triangleq \frac{1}{\sqrt{2\pi}}e^{j\Omega t}$ and basis coefficients $\frac{1}{\sqrt{2\pi}}X_{\mathsf{c}}(j\Omega)$:

$$x(t) = \int_{-\infty}^{\infty} \tfrac{1}{\sqrt{2\pi}}X_{\mathsf{c}}(j\Omega)\, b_\Omega(t)\,\mathrm{d}\Omega$$

$$\tfrac{1}{\sqrt{2\pi}}X_{\mathsf{c}}(j\Omega) = \int_{-\infty}^{\infty} x(t)\, b_\Omega^*(t)\,\mathrm{d}t,$$

  where $\frac{1}{\sqrt{2\pi}}X_{\mathsf{c}}(j\Omega)$ tells how much $b_\Omega(t)$ is present in $x(t)$.

- When $\{x(t)\}_{t\in\mathbb{R}}$ is a sum of a few sinuoids, the CTFT $\{X_{\mathsf{c}}(j\Omega)\}_{\Omega\in\mathbb{R}}$ is very nice because it gives us a concise description of how the signal is constructed:



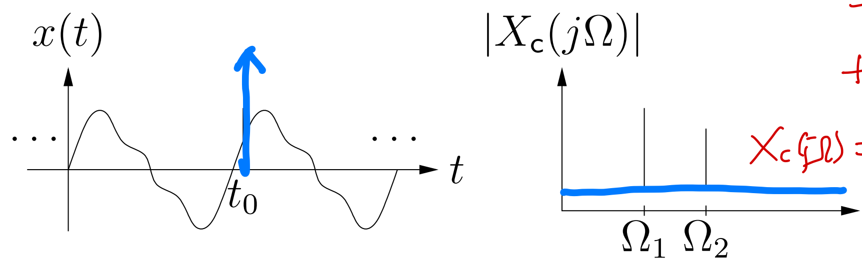
- But other times, the CTFT does not give the best "view."

Limitations of Fourier Analysis (cont.):

- For example, consider adding a sharp "glitch" (e.g., a Dirac delta) to the sum-of-sinusoids signal at time t_0. This adds $\{e^{-j\Omega t_0}\}_{\Omega\in\mathbb{R}}$ to the CTFT:

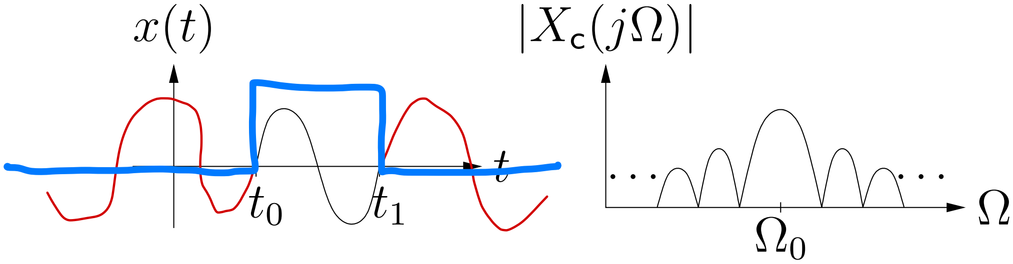
$x(t) = a_1 \cos(\Omega_1 t) + a_2 \cos(\Omega_2 t) + a_3 \delta(t-t_0)$

$X_c(j\Omega) = a_1 \pi[\delta(\Omega-\Omega_1)+\delta(\Omega+\Omega_1)] + a_2\pi[\delta(\Omega-\Omega_2)+\delta(\Omega+\Omega_2)] + a_3 e^{j\Omega t_0}$



The glitch info t_0 is better visible in the time domain.

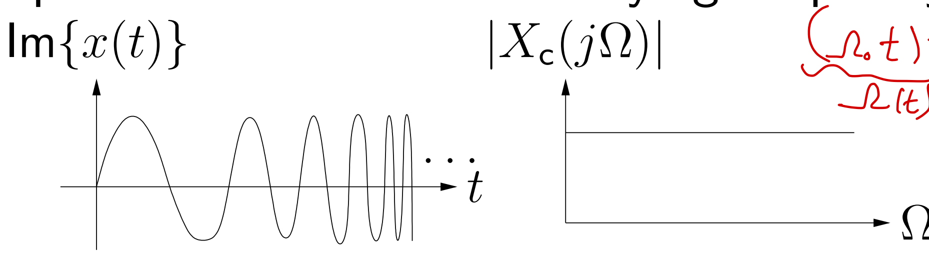
- Or consider a windowed version of $\{\sin(\Omega_0 t)\}_{t\in\mathbb{R}}$:



Although $x(t)$ consists of a "pure" sinusoid over $t \in [t_0, t_1]$, this is not clearly visible from the CTFT.

- Or consider the complex linear chirp $x(t) = e^{j\Omega_0 t^2} = e^{j\Omega(t)t}$, a complex sinusoid with time-varying frequency $\Omega(t) = \Omega_0 t$

$\underbrace{(\Omega_0 t)}_{\Omega(t)} t$



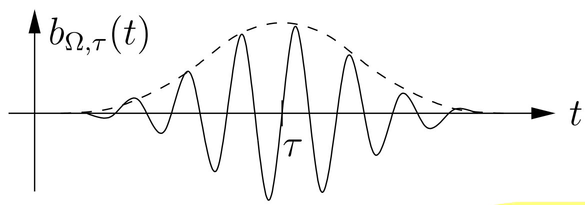
The CTFT magnitude is flat; it doesn't show the interesting structure of the chirp signal.

The Short-Time CTFT:

- The problem with the CTFT is that "frequency" is *time-invariant*. That is, the CTFT constructs $x(t)$ from sinusoids $b_\Omega(t) = e^{j\Omega t}$ spread uniformly over all time t.

- What if we instead constructed a basis from *windowed sinusoids* localized at time $t \approx \tau$ (for many different τ):

$$b_{\Omega,\tau}(t) \triangleq \frac{1}{\sqrt{2\pi}} w(t-\tau) e^{j\Omega t} \text{ for } t \in \mathbb{R},$$



We'll assume real-valued $w(t)$ s.t. $\int_{-\infty}^{\infty} w(t)^2 \, \mathrm{d}t = 1$.

- We could use them to construct/deconstruct $x(t)$ via

$$x(t) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} X_{\mathsf{c}}(j\Omega, \tau) \, b_{\Omega,\tau}(t) \, \mathrm{d}\Omega \, \mathrm{d}\tau$$

$$\frac{1}{\sqrt{2\pi}} X_{\mathsf{c}}(j\Omega, \tau) = \underbrace{\int_{-\infty}^{\infty} x(t) b^*_{\Omega,\tau}(t) \, \mathrm{d}t}_{b_{\Omega,\tau}(t) \cdot x(t)} \quad \leftarrow \text{"coefficients"}$$

- After rearranging, we have the *short-time CTFT*:

$$X_{\mathsf{c}}(j\Omega, \tau) = \int_{-\infty}^{\infty} x(t) w(t-\tau) e^{-j\Omega t} \, \mathrm{d}t$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} X_{\mathsf{c}}(j\Omega, \tau) w(t-\tau) e^{j\Omega t} \, \mathrm{d}\Omega \, \mathrm{d}\tau.$$

$$X_c(j\Omega, z) = \int_{-\infty}^{\infty} \underbrace{x(t)\, w(t-z)}_{\tilde{x}(t)} e^{-j\Omega t}\, dt$$

$$= w(t-z)e^{j\Omega t} \cdot x(t)$$

windowed FT

$$= e^{j\Omega t} \cdot \tilde{x}(t)$$

$$\tilde{x}(t) = x(t)\, w(t-z) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X_c(j\Omega, z) e^{j\Omega t}\, d\Omega$$

$$x(t)\, w(t-z)\, w(t-z) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X_c(j\Omega, z)\, w(t-z)\, e^{j\Omega t}\, d\Omega$$

$$\int_{-\infty}^{\infty} x(t)\, w^2(t-z)\, dz = \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} X_c(j\Omega, z)\, w(t-z)\, e^{j\Omega t}\, d\Omega\, dz$$

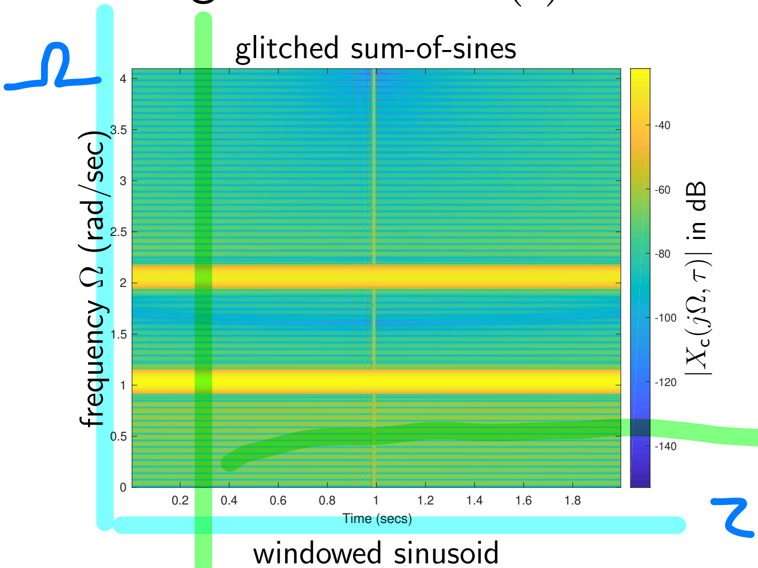
For any fixed t, Let $t-z=s$

$$\int_{-\infty}^{\infty} x(t)\, w^2(t-z)\, dz = \int_{-\infty}^{\infty} x(t)\, w^2(s)\, ds = x(t)\underbrace{\int_{-\infty}^{\infty} w^2(s)\, ds}$$

ST-CTFT of Previous Examples:

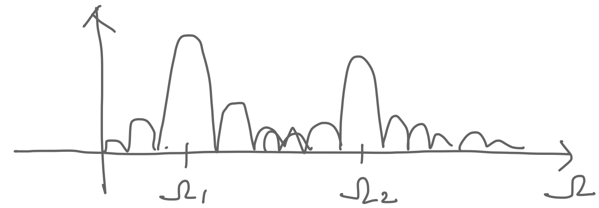
Using a window $w(t)$ of width $1/64$ sec...

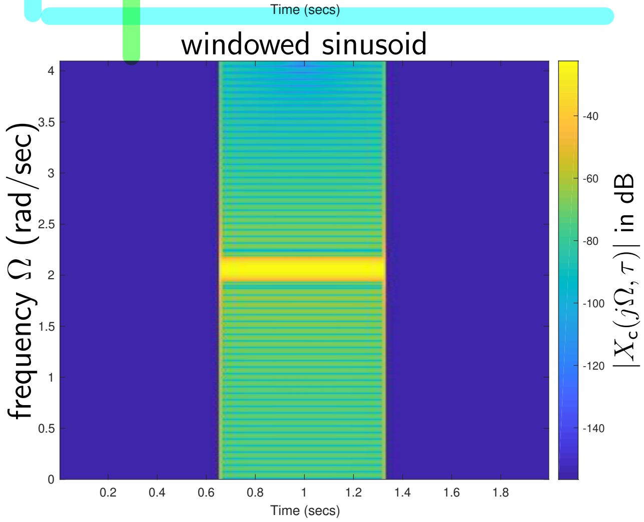
$x(t) = a_1 \cos(\Omega_1 t) + a_2 \cos(\Omega_2 t)$
$+ a_3 \delta(t - t_0)$



Note sinusoids at $\Omega_1 = 1$ and $\Omega_2 = 2$, and glitch at time $\tau = 1$.

$t_0 = 1$

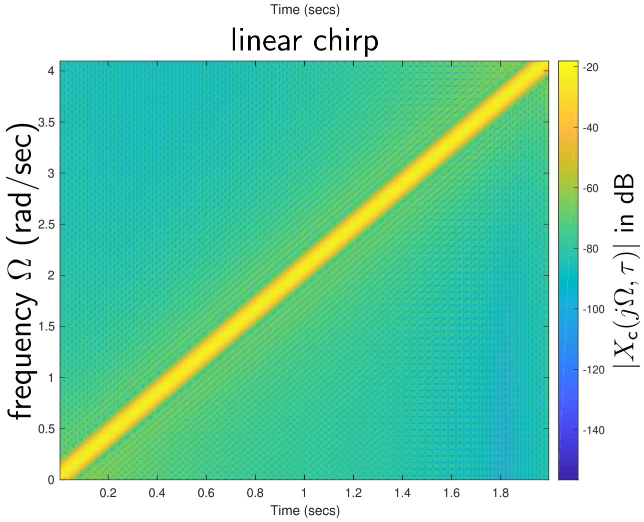




Note sinusoid at $\Omega_0 = 2$ windowed over time interval $\tau \in [\frac{2}{3}, \frac{4}{3}]$.

$x(t) = w(t - t_0) e^{j\Omega_0 t}$

$x(t) = e^{j(\Omega t)t}$

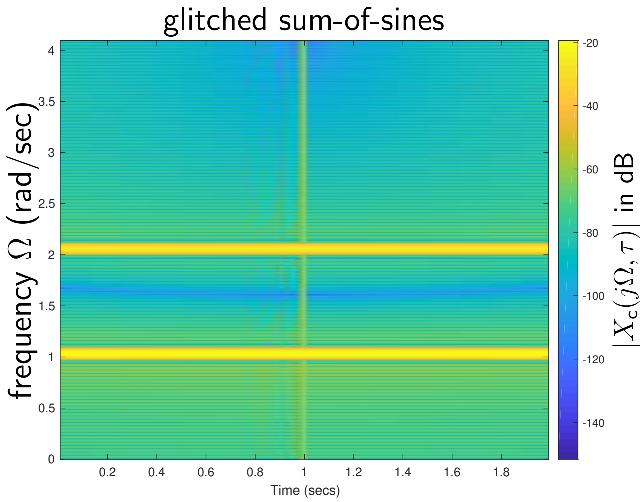


Note linear chirp with instantaneous frequency $\Omega(\tau) = \tau$.

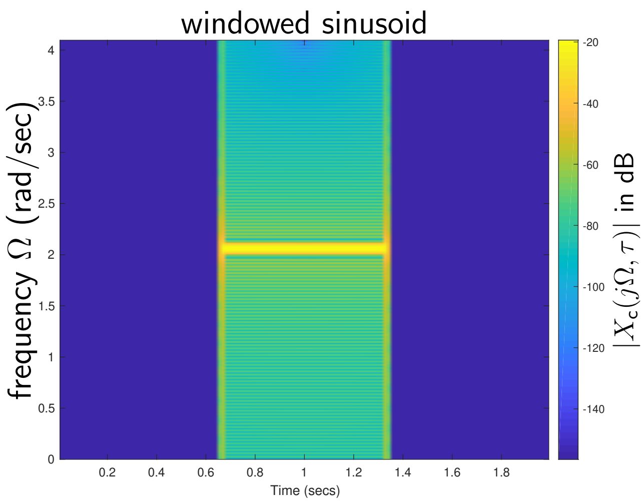
Note resolutions in time and frequency domains.

ST-CTFT of Previous Examples (cont.):

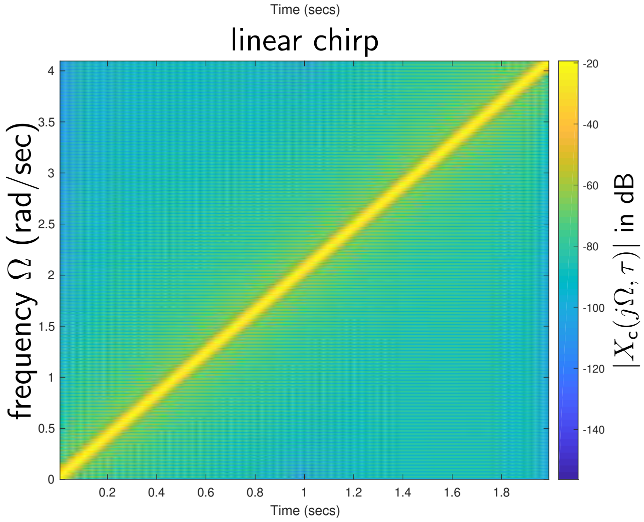
Using a wider window $w(t)$ of width $1/32$ sec...



Note sinusoids are sharper but glitch is more blurred.



Note sinusoid is sharper but edges are more blurred.



Similar to before.

Frequency resolution improved, but time resolution degraded!

$p(x)$: probability density function $\qquad$ mean: $\mu = \int x\, p(x)\, dx$

$p(x) \geq 0, \quad \int p(x)\, dx = 1 \qquad \sigma^2 = \int (x-\mu)^2 p(x)\, dx$

Time and Frequency Resolution:

- The previous examples showed that, as the window $w(t)$ gets wider, the ST-CTFT frequency resolution improves but the time resolution degrades.
- Let's try to understand this more quantitatively.
- Consider an arbitrary basis element $b(t)$ with energy

$$E \triangleq \int_{-\infty}^{\infty} |b(t)|^2 \, \mathrm{d}t. \qquad \int_{-\infty}^{\infty} |B_c(j\Omega)|^2 d\Omega = 2\pi E$$

- Let's define the time/frequency *centers* and *widths*

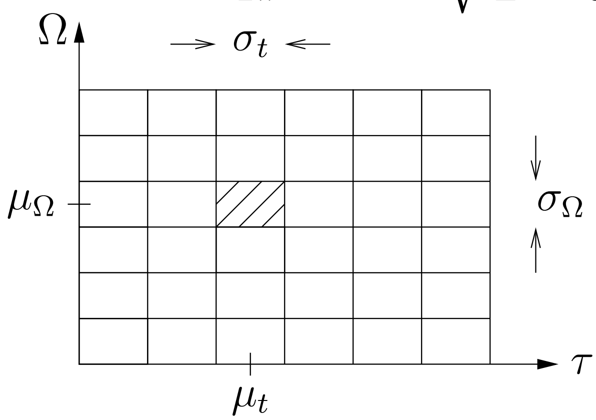
mean $$\mu_t \triangleq \frac{1}{E}\int_{-\infty}^{\infty} t\,|b(t)|^2 \, \mathrm{d}t = \int t\,p(t)\,dt \quad \sigma_t \triangleq \sqrt{\frac{1}{E}\int_{-\infty}^{\infty} (t-\mu_t)^2\,|b(t)|^2 \, \mathrm{d}t}$$

$$\mu_\Omega \triangleq \frac{1}{E}\int_{-\infty}^{\infty} \Omega\,|B_{\mathsf{c}}(j\Omega)|^2 \frac{\mathrm{d}\Omega}{2\pi}, \quad \sigma_\Omega \triangleq \sqrt{\frac{1}{E}\int_{-\infty}^{\infty} (\Omega-\mu_\Omega)^2\,|B_{\mathsf{c}}(j\Omega)|^2 \frac{\mathrm{d}\Omega}{2\pi}}$$

$p(t) = \dfrac{|b(t)|^2}{E}$

$p(t) \geq 0$

$\int p(t)\,dt = \frac{1}{E}\int |b(t)|^2 dt = 1$



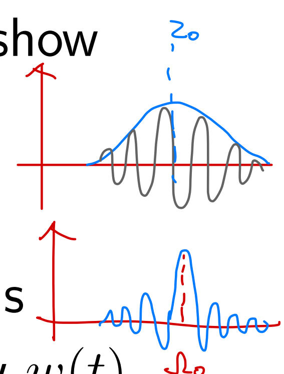
$q(\Omega) = \dfrac{|B_c(j\Omega)|^2}{2\pi E}$

$q(\Omega) \geq 0$

$\int q(\Omega)\, d\Omega = 1$

- For ST-CTFT basis element $b_{\Omega_0,\tau_0}(t) = \frac{1}{\sqrt{2\pi}} w(t-\tau_0)e^{j\Omega_0 t}$ with real, symmetric, unit-energy $w(t)$, one can show

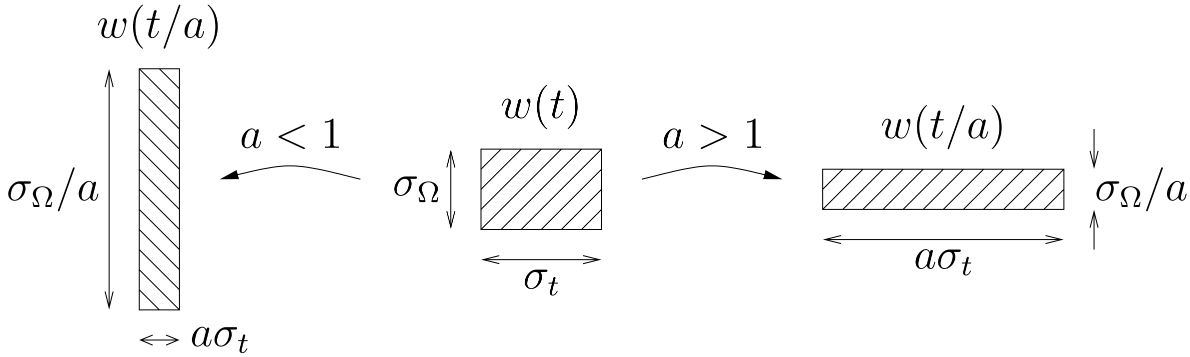
$$\mu_t = \tau_0, \qquad \sigma_t = \sqrt{\int_{-\infty}^{\infty} t^2\,|w(t)|^2\,\mathrm{d}t}$$
$$\mu_\Omega = \Omega_0, \quad \sigma_\Omega = \sqrt{\frac{1}{2\pi}\int_{-\infty}^{\infty} \Omega^2\,|W_{\mathsf{c}}(j\Omega)|^2\,\mathrm{d}\Omega}.$$



So $b_{\Omega_0,\tau_0}(t)$ is centered at (τ_0, Ω_0) and has widths $(\sigma_t, \sigma_\Omega)$ that depend on the shape of the window $w(t)$.

Time/Frequency Uncertainty Principle:

- Stretching $w(t)$ in one domain compresses it in the other:



  implying a tradeoff between time & frequency resolution.

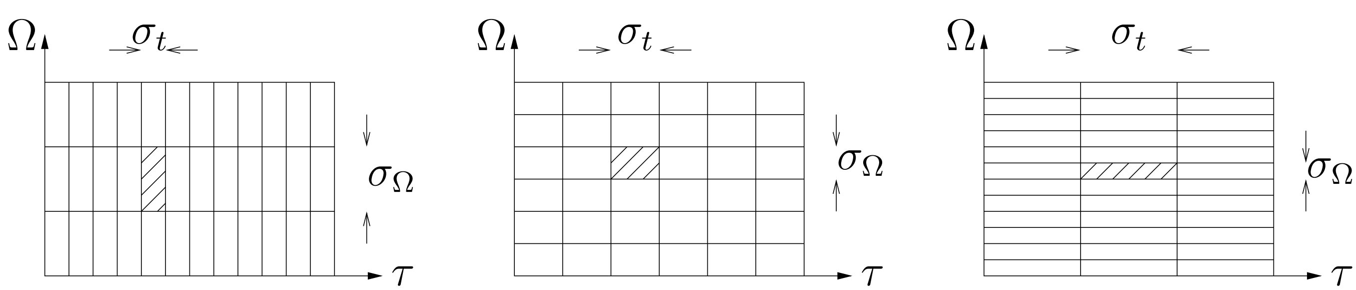
- The *time-frequency uncertainty principle* says, for any window $w(t)$, and thus any $b(t)$,

$$\sigma_t \sigma_\Omega \geq \tfrac{1}{2},$$

  where equality is reached by a Gaussian window, i.e.,

$$w(t) = (2c/\pi)^{\frac{1}{4}} e^{-ct^2} \text{ for any } c > 0.$$

- Together, the ST-CTFT basis functions $\{b_{\Omega,\tau}(t)\}$ form a "tiling" of the time-frequency plane that looks like



  depending on the window $w(t)$ used to construct $b_{\Omega,\tau}(t)$. The tile *area* can be no smaller than $1/2$.

Proof of Uncertainty Principle

We assume $x(t) \in \mathbb{R}$ for all $t \in \mathbb{R}$. We also assume $\int_{-\infty}^{\infty} |x(t)|^2 dt < \infty$. Then it follows that $\lim_{t\to\infty} x(t) = 0$ and $\lim_{t\to-\infty} x(t) = 0$.

Let us consider

$(fg)' = f'g + fg'$ $\qquad \int fg' = fg - \int f'g$

$$-\int_{-\infty}^{\infty} t x(t) x'(t) dt = -\underbrace{\frac{t[x(t)]^2}{2}\Bigg|_{-\infty}^{\infty}}_{=0} + \underbrace{\int_{-\infty}^{\infty} \frac{[x(t)]^2}{2} dt}_{E/2}$$

$f(t) = t$

$g'(t) = x(t)\, x'(t)$

$$= \frac{1}{2}\sqrt{\underbrace{\int_{-\infty}^{\infty} |x(t)|^2 dt}_{E}} \cdot \sqrt{\underbrace{\frac{1}{2\pi}\int_{-\infty}^{\infty} |X_c(j\Omega)|^2 d\omega}_{E}}.$$

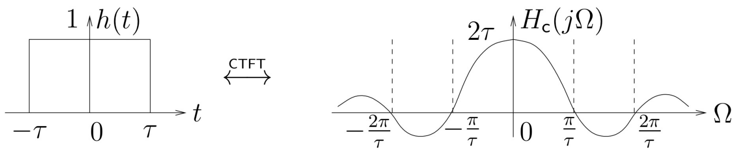
On the other hand, by the Cauchy-Schwarz inequality, we have

$$\underbrace{\left|\int_{-\infty}^{\infty} \underbrace{t x(t)}_{f(t)}\, \underbrace{x'(t)}_{g(t)} dt\right|}_{f(t)\,\cdot\, g(t)} \le \sqrt{\int_{-\infty}^{\infty} |t x(t)|^2 dt} \cdot \sqrt{\int_{-\infty}^{\infty} |x'(t)|^2 dt}$$

$$= \sqrt{\int_{-\infty}^{\infty} t^2 |x(t)|^2 dt} \cdot \sqrt{\frac{1}{2\pi}\int_{-\infty}^{\infty} \Omega^2 |X_c(j\Omega)|^2 d\omega}.$$

3. Rectangle:

$$\begin{cases} 1 & \text{if } |t| < \tau \\ 0 & \text{else} \end{cases} \quad \overset{\text{CTFT}}{\longleftrightarrow} \quad 2\frac{\sin(\Omega\tau)}{\Omega}$$



$$\frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{t^2}{2\sigma^2}} \quad \overset{\text{CTFT}}{\longleftrightarrow} \quad e^{-\frac{\sigma^2\Omega^2}{2}}$$

Short-time DTFT:

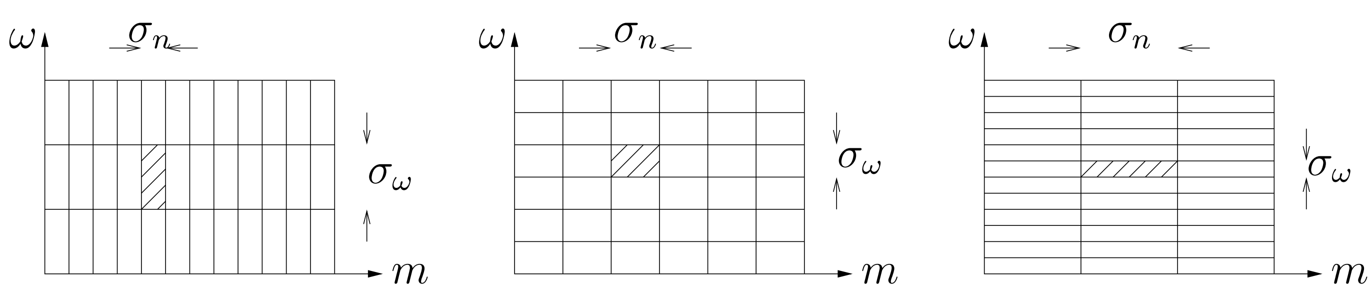
- The ST-CTFT applies to continuous-time signals $x(t)$.
- For discrete-time signals $x[n]$, we have the ST-DTFT:

$$X(e^{j\omega}, m) = \sum_{n=-\infty}^{\infty} x[n]w[n-m]e^{-j\omega n}$$

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi}\sum_{m=-\infty}^{\infty} X(e^{j\omega}, m)w[n-m]e^{j\omega n}\,\mathrm{d}\omega,$$

  where we have assumed $w[n] \in \mathbb{R}$ and $\sum_n w[n]^2 = 1$.

- The ST-DTFT constructs/deconstructs $x[n]$ using basis sequences $b_{\omega,m}[n] = \frac{1}{\sqrt{2\pi}}w[n-m]e^{j\omega n}$ for various ω, m.
- The ST-DTFT is very similar to the ST-CTFT, except that the time and shift domains are discrete and the frequency domain is 2π-periodic.



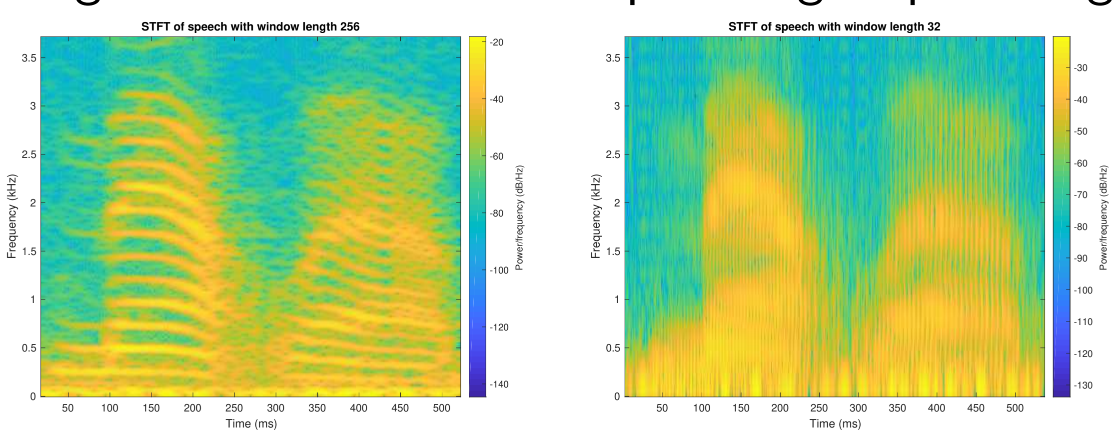
The definitions of σ_n and σ_ω in the figures are similar to those of σ_t and σ_Ω from earlier.

Short-time DFT:

- Most commonly, the ST-DTFT is used for *analysis*: we want to compute $|X(e^{j\omega}, m)|$ but we don't care about reconstructing $x[n]$ from $X(e^{j\omega}, m)$. (We'll talk about reconstruction later, in the context of filterbanks.)
- For practicality, ω is sampled at N frequencies $\{\frac{2\pi k}{N}\}_{k=0}^{N-1}$ and m is downsampled by some factor $M \geq 1$:

$$X[k, l] \triangleq X(e^{j\frac{2\pi}{N}k}, lM) = \sum_{n=-\infty}^{\infty} x[n]w[n - lM]e^{-j\frac{2\pi}{N}kn},$$

  giving the ST-DFT.
- As before, the length of the window $\{w[n]\}$ determines the time resolution and frequency resolution.
- Matlab's "`spectrogram`" computes & plots the ST-DFT magnitude. Here's an example using a speech signal:



Left plot shows formant at $\approx$ 200 Hz.

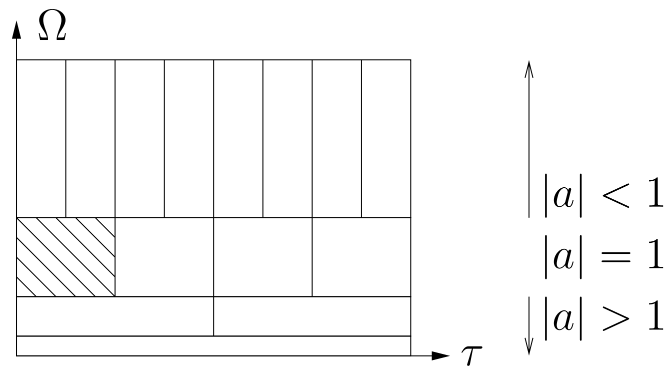
Right plot shows 5 ms periodicity, which is consistent.

  - With a long window (left), we can see individual harmonics.
  - With a short window (right), we can see individual oscillations.

A Better Tiling?:

- The ST-CTFT uses *time & frequency shifts* of a single window function: $b_{\Omega,\tau}(t) = \frac{1}{\sqrt{2\pi}} w(t-\tau) e^{j\Omega t}$.
  - The choice of the window $w(t)$ involves a *tradeoff* between time & frequency resolution.
- What if we instead used *time shifts and stretches* of a single function $\psi(t)$? This would yield a tiling like

$x(t) = \int_{-\infty}^{\infty} x(z)\, \delta(t-z)\, dz$



Good time resolution *and* good frequency resolution!

- To understand why this latter tiling may be better. . .
  - Say we wanted to distinguish between sinusoids at frequencies 1000 & 1001 Hz. After 1 sec, they differ by 1 period $\Rightarrow$ orthogonal $\Rightarrow$ easy to tell apart.
  - Now consider sinusoids at freqs 1 Hz & 1.001 Hz. After *1000 sec*, they differ by 1 period $\Rightarrow$ orthogonal $\Rightarrow$ easy to tell apart.
  - So, as the frequency Ω *decreases*, we require a *longer* observation period for good frequency resolution.
  - The above tiling does this!
  - It also provides good time resolution via the skinny top tiles!

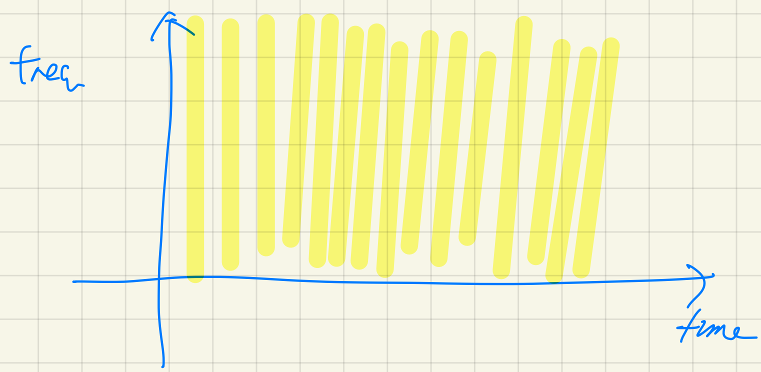
$x(t)$: CT signal

$$X_{\Omega,\tau} = \int_{-\infty}^{\infty} x(t)\, b^*_{\Omega,\tau}(t)\, dt$$

freq   shift

ex1) $b_{\Omega,\tau}(t) = e^{j\Omega t}$ (indep of τ)

ex2) $b_{\Omega,\tau}(t) = \delta(t-\tau)$



$e^{j2\pi \cdot 1000t}$ $e^{j2\pi 1001t}$

1000 Hz 1001 Hz

observe for 1 sec

$$\int_0^1 e^{-j2\pi 1000t} e^{j2\pi 1001t} dt$$

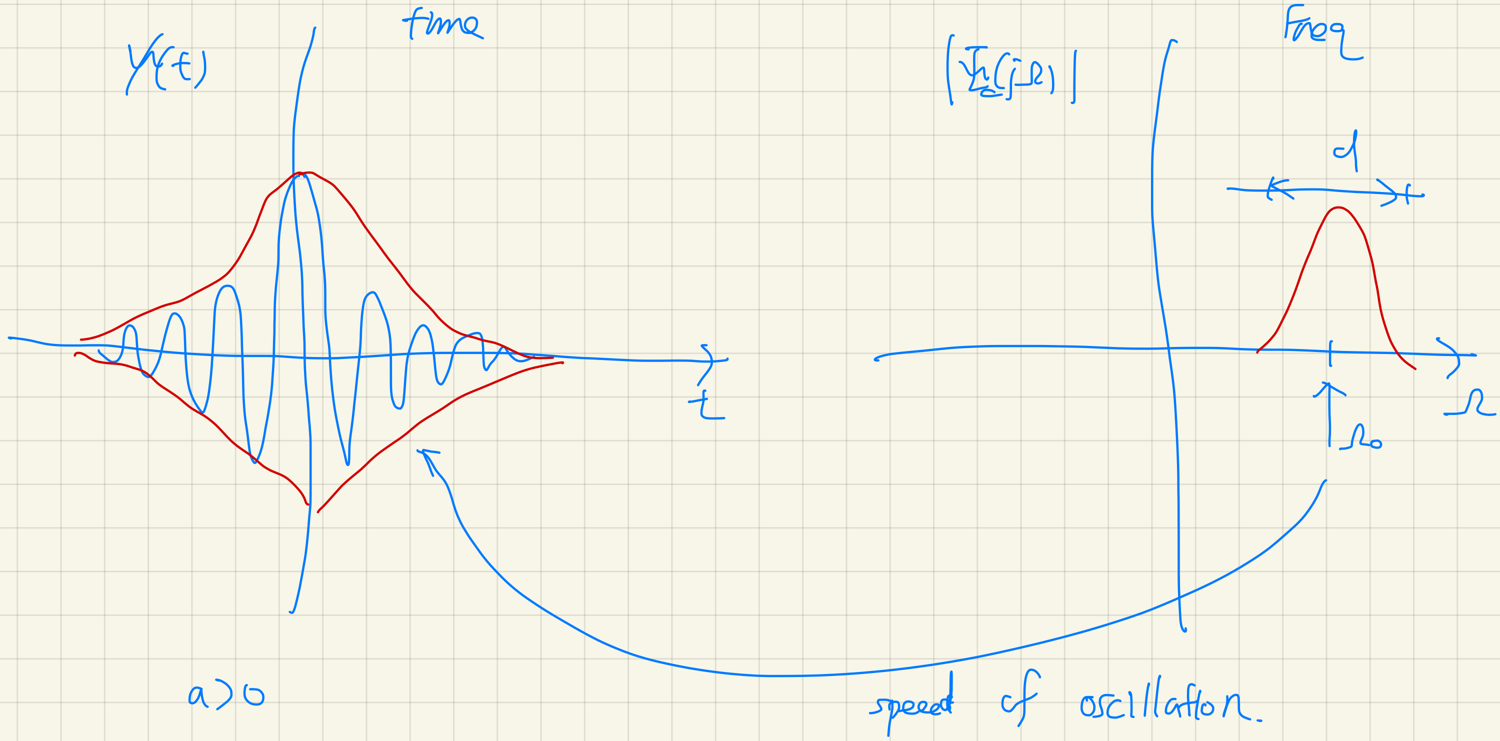
$$= \int_0^1 e^{j2\pi t} dt = 0$$

$e^{j2\pi t}$ $e^{j2\pi (1.001)t}$

1 Hz 1.001 Hz

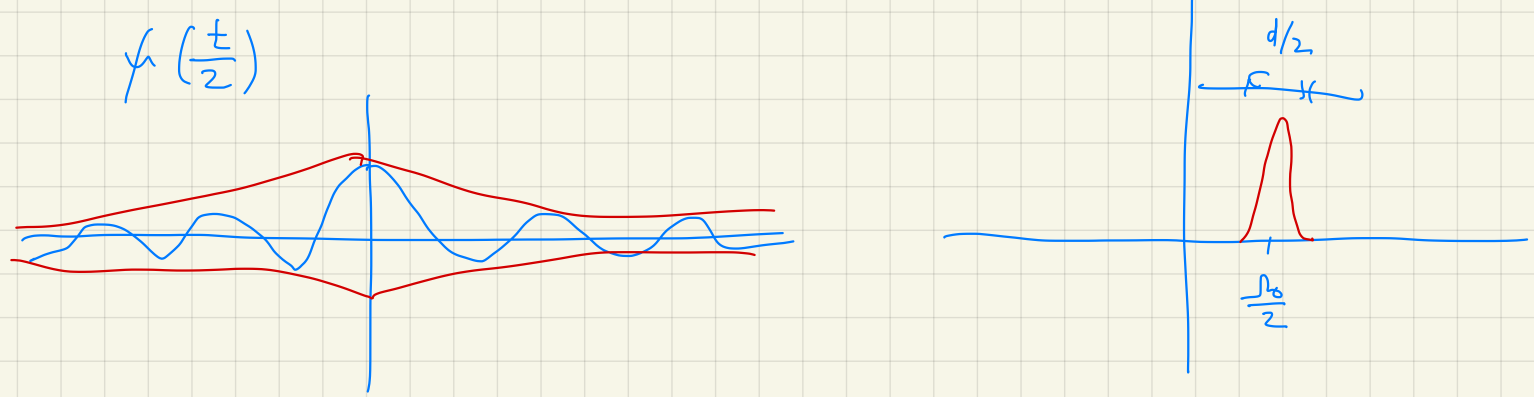
$$\int_0^{1000} e^{-j2\pi t} e^{j2\pi (1.001t)} dt$$

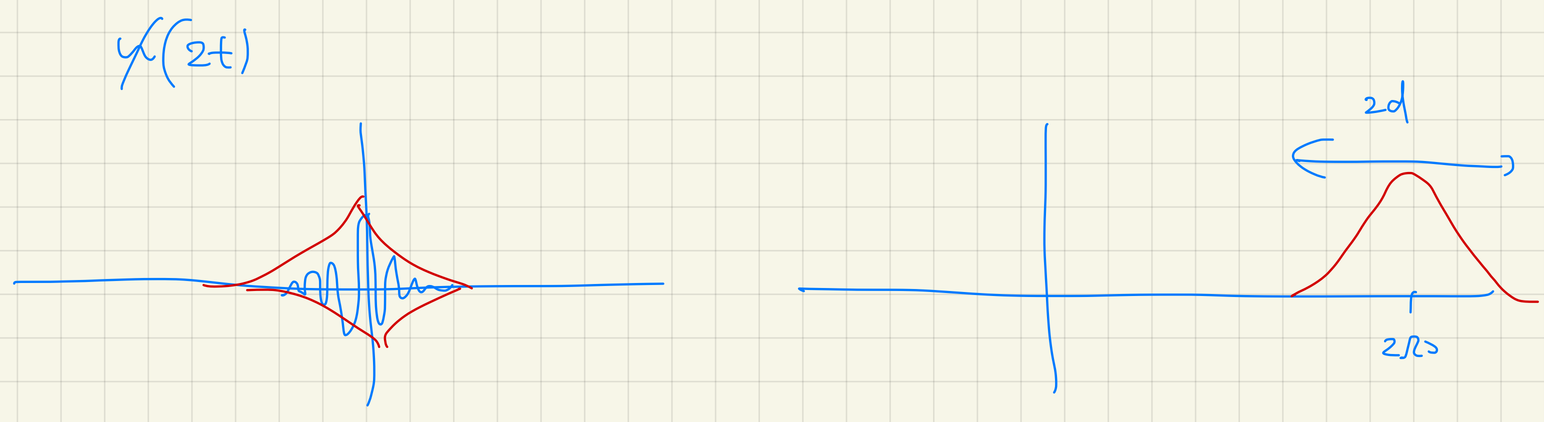
$$= \int_0^{1000} e^{j2\pi (0.001t)} dt = 0$$



$a>0$

$$x\left(\frac{t}{a}\right) \xleftrightarrow{FT} a\, X_c(ja\Omega)$$





The Continuous Wavelet Transform:

- The tiling on the previous page results from the *continuous wavelet transform* (CWT)

$$X_{\text{cwt}}(a,\tau) = \int_{-\infty}^{\infty} x(t)\psi^*_{a,\tau}(t)\,\mathrm{d}t, \qquad \tau \in \mathbb{R}, a > 0$$

$$x(t) = \frac{1}{C_\psi}\int_{0_+}^{\infty}\int_{-\infty}^{\infty} X_{\text{cwt}}(a,\tau)\psi_{a,\tau}(t)\frac{\mathrm{d}\tau\,\mathrm{d}a}{a^2}$$

where $\psi_{a,\tau}(t)$ is an a-stretched and τ-shifted copy of some "mother wavelet" $\psi(t)$:

dilation

$\int_{-\infty}^{\infty} f(t)e^{-j0t}\,dt = \Psi_c(j0) = 0$

$$\boxed{\psi_{a,\tau}(t) = \frac{1}{\sqrt{|a|}}\psi\left(\frac{t-\tau}{a}\right)} \text{ with } \begin{cases} \int_{-\infty}^{\infty}\psi(t)\,\mathrm{d}t = 0 \\ C_\psi \triangleq \int_{-\infty}^{\infty}\frac{|\Psi_c(j\Omega)|^2}{|\Omega|}\mathrm{d}\Omega < \infty \end{cases}$$

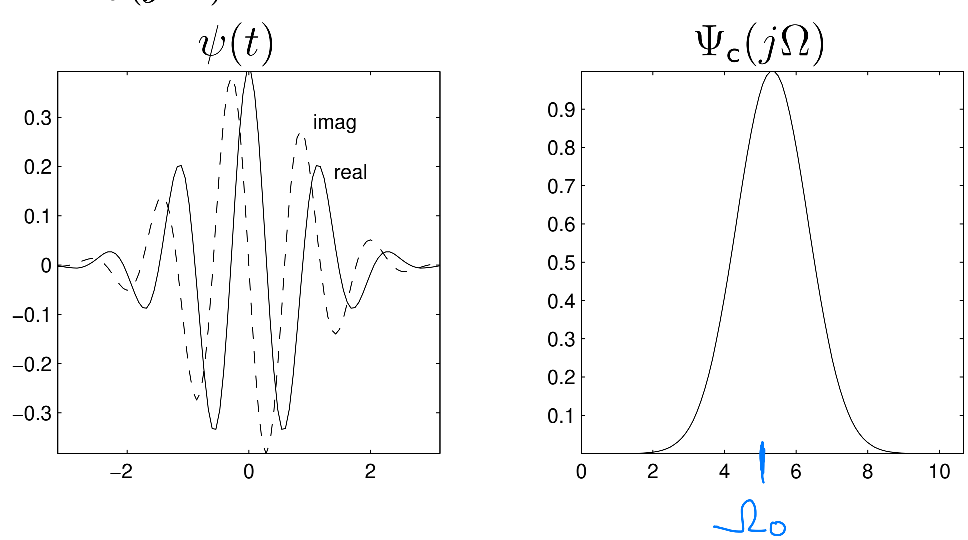
- A classic example is the Morlet wavelet, which uses a modulated Gaussian pulse:

oscillation

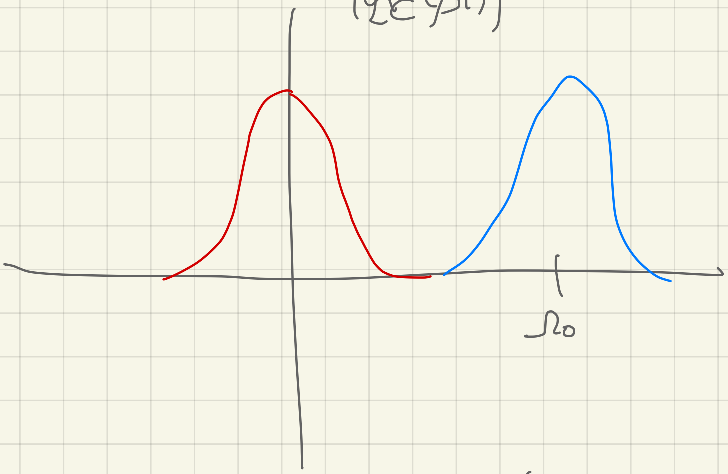
$$\psi(t) = \frac{1}{\sqrt{2\pi}}e^{j\Omega_0 t}e^{-t^2/2}, \quad \Omega_0 = \pi\sqrt{\frac{2}{\ln 2}}$$

envelope

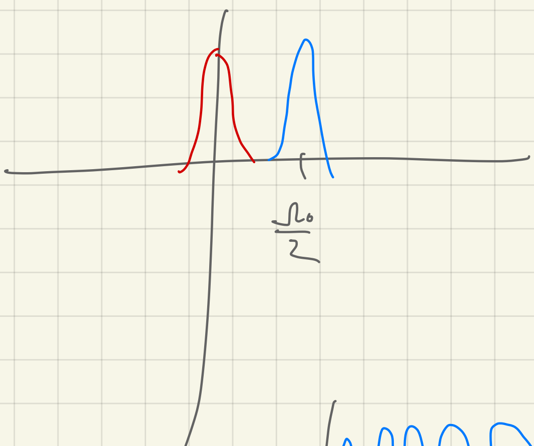
$$\Psi_c(j\Omega) = e^{-(\Omega-\Omega_0)^2/2}$$



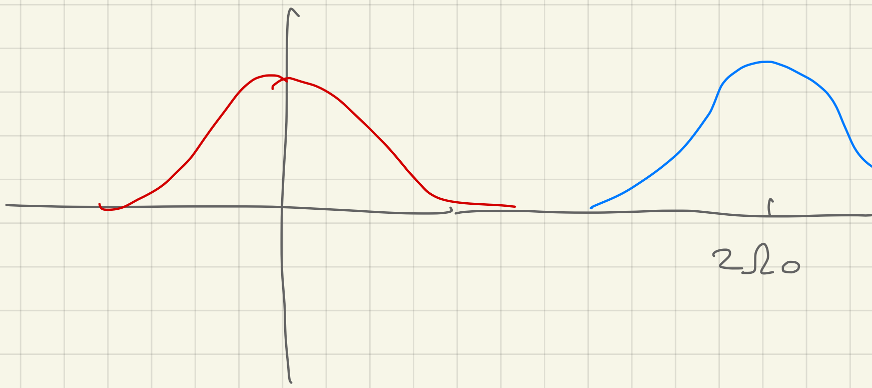
$|X_c(j\Omega)|$

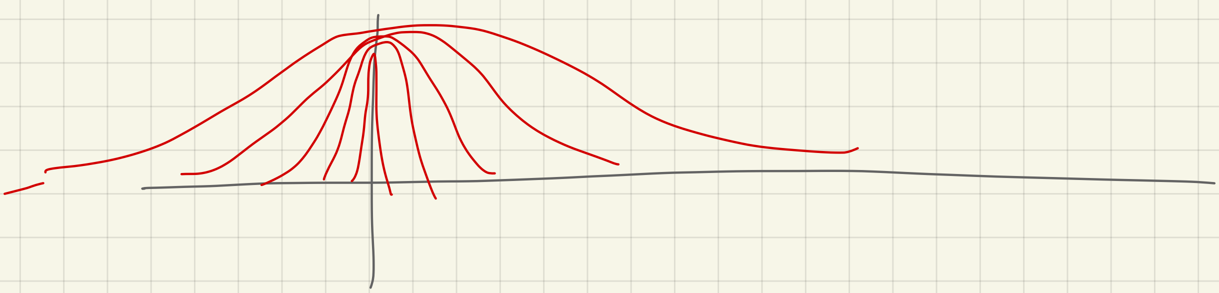


$|X_c(j2\Omega)|$



$|X_c(j\frac{\Omega}{2})|$





conjugate signal

given $f(t)$, let $\tilde{f}(t) = f^*(-t)$ $\quad$ $\tilde{f}$: conjugate of f

$$\Updownarrow \text{FT}$$

$$\tilde{F}_c(j\Omega) = F^*(j\Omega)$$

$$W(a,z) = \int_{-\infty}^{\infty} x(t)\, \psi_{a,z}^*(t)\, dt$$

$$= \int_{-\infty}^{\infty} x(t) \frac{1}{\sqrt{a}} \psi^*\left(\frac{t-z}{a}\right) dt$$

$$\psi^*\left(\frac{t-z}{a}\right) = \tilde{\psi}\left(\frac{z-t}{a}\right)$$

$$= \int_{-\infty}^{\infty} x(t) \frac{1}{\sqrt{a}} \tilde{\psi}\left(\frac{z-t}{a}\right) dt$$

$$= x(t) * \tilde{\psi}_{a,z}(t) \Big|_{t=z}$$

$$\int_{0^+}^{\infty} \left(\underbrace{\int_{-\infty}^{\infty} W(a,z)\, \psi_{a,z}(t)\, dz}_{\int_{-\infty}^{\infty} W(a,z) \frac{1}{\sqrt{a}} \psi\left(\frac{t-z}{a}\right) dz} \right) \frac{da}{a^2}$$

$$= x(t) * \tilde{\psi}_{a,z}(t) * \psi_{a,z}(t)$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\Omega) \left| \Psi_{a,z}(j\Omega) \right|^2 e^{j\Omega t}\, d\Omega$$

$$\int_{0^+}^{\infty}\left[\int_{-\infty}^{\infty} W(a,\tau)\,\psi_{a,\tau}(t)\,d\tau\right]\frac{da}{a^2}$$

$$= \int_{0^+}^{\infty}\frac{1}{2\pi}\int_{-\infty} X_c(j\Omega)\,|\Psi_{a,\tau}(j\Omega)|^2 e^{j\Omega t}\,d\Omega\,\frac{da}{a^2}$$

$$\psi_{a,\tau}(t) = \frac{1}{\sqrt{a}}\,\psi\left(\frac{t-\tau}{a}\right)$$

$$\updownarrow$$

$$|\Psi_{a,\tau}(j\Omega)| = |\sqrt{a}\,\Psi_c(ja\Omega)\,e^{-j\Omega\tau}|$$

$$= \sqrt{a}\,|\Psi_c(ja\Omega)|$$

$$= \int_{0^+}^{\infty}\frac{1}{2\pi}\int_{-\infty}^{\infty} X_c(j\Omega)\,\frac{|\Psi_c(ja\Omega)|^2}{a}\,e^{j\Omega t}\,d\Omega\,da$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} X_c(j\Omega)e^{j\Omega t}\boxed{\left[\int_{0^+}^{\infty}\frac{|\Psi_c(ja\Omega)|^2}{a}\,da\right]}\,d\Omega$$

let Ω fixed

apply change of variable let $b = \Omega a$

$db = \Omega\, da$

$$\boxed{} = \int_{0^+}^{\infty}\frac{|\Psi_c(jb)|^2}{b}\,db = C_\psi$$

$$\hat{x}(t) = C_\psi\,\underbrace{\frac{1}{2\pi}\int_{-\infty}^{\infty} X_c(j\Omega)\,e^{j\Omega t}\,d\Omega}_{x(t)}$$

$$\psi(t) = \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} \left(e^{j\Omega_0 t} - K_{\Omega_0}\right)$$

$$K_{\Omega_0} = e^{-\frac{1}{2}\Omega_0^2}$$

$$\Psi_c(j\Omega) = e^{-\frac{1}{2}(\Omega-\Omega_0)^2} - K_{\Omega_0} e^{-\frac{1}{2}\Omega^2}$$

$$\Omega_0 = \pi \sqrt{\frac{2}{\ln 2}}$$

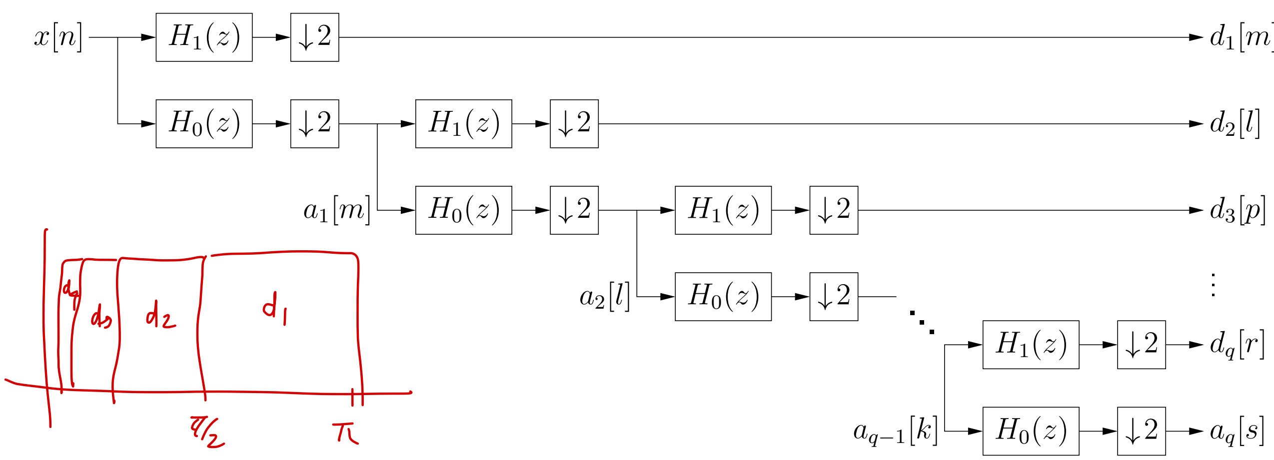
$$K_{\Omega_0} = e^{-\frac{\pi^2}{\ln 2}} \approx 6.54 \times 10^{-7}$$

The Discrete Wavelet Transform (DWT):

- The CWT is not very practical since the time t, stretch a, and shift τ variables are all *continuous valued*.

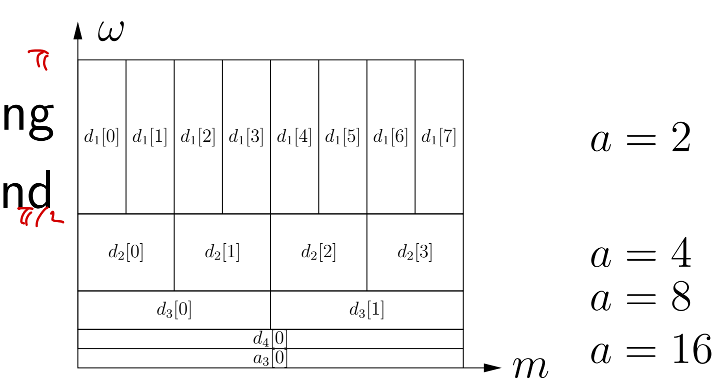
- To be practical, we'd like a wavelet transform that operates on *discrete-time* signals $x[n]$ and uses *discrete-valued* stretches and shifts, i.e., a DWT.

- We can do this by restricting to "dyadic" stretches $a \in \{2^q : q \in \mathbb{Z}_+\}$ and using an iterated filterbank

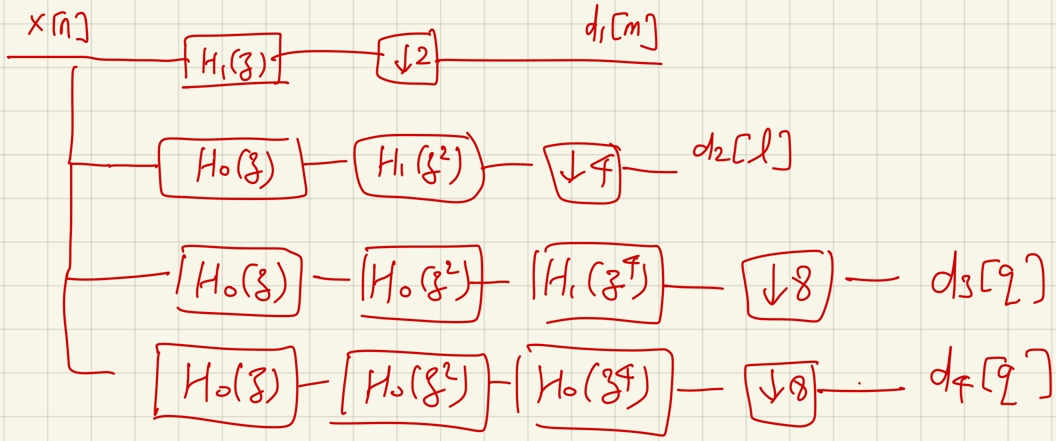


  with appropriate highpass $H_1(z)$ and lowpass $H_0(z)$.

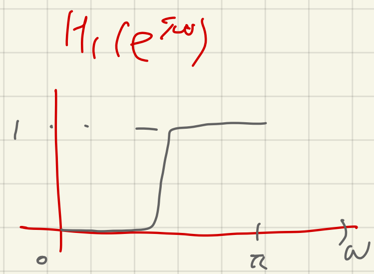
- This yields the tiling (for $Q = 4$ levels and length $N = 16$):



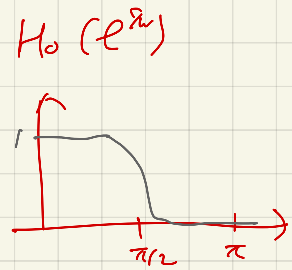
- Note that the # DWT outputs = # DWT inputs!



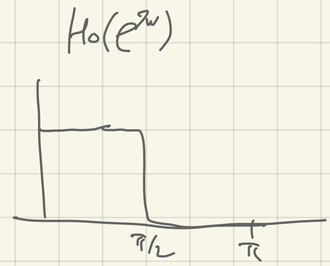
x[n]
H1(z)
↓2
d1[m]
H0(z)
H1(z²)
↓4
d2[l]
H0(z²)
H1(z⁴)
↓8
d3[q]
H0(z⁴)
d4[q]



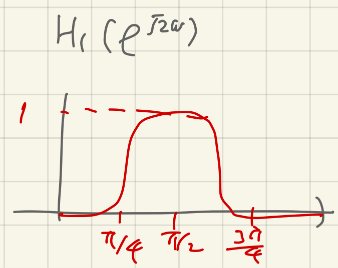
H1(e^jω)
1
0
π
ω



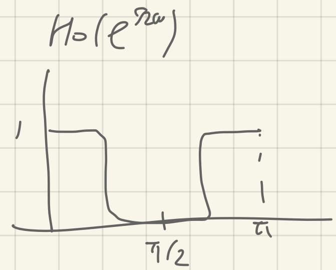
H0(e^jω)
1
π/2
π



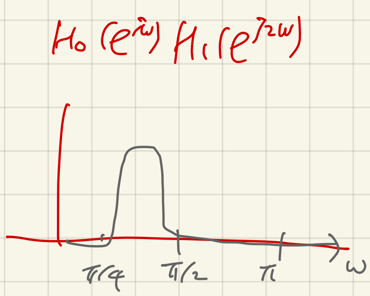
H0(e^jω)
π/2
π



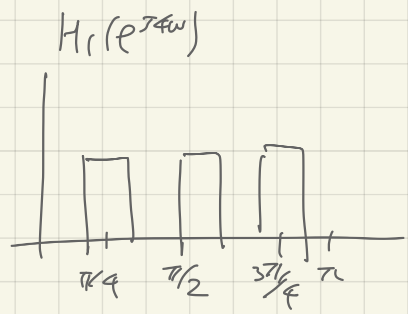
H1(e^j2ω)
1
π/4
π/2
3π/4



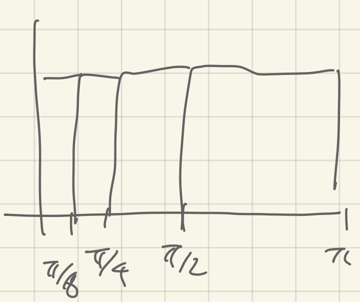
H0(e^j2ω)
1
π/2
π



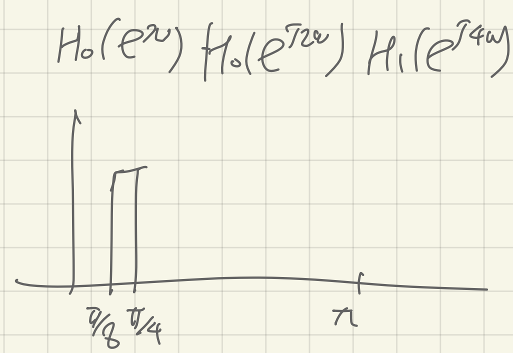
H0(e^jω) H1(e^j2ω)
π/4
π/2
π
ω



H1(e^j4ω)
π/4
π/2
3π/4
π



π/8
π/4
π/2
π



H0(e^jω) H0(e^j2ω) H1(e^j4ω)
π/8
π/4
π

$$\tilde{g}[n] = g^*[-n] \qquad \tilde{G}(e^{j\omega}) = G^*(e^{j\omega})$$

conjugate signal

$$x[n] \longrightarrow \boxed{H_1(z)} \xrightarrow{v[n]} \boxed{\downarrow 2} \longrightarrow d_1[m]$$

$$v[n] = \sum_{k=-\infty}^{\infty} x[k]\, h_1[n-k]$$

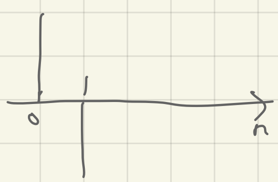
$$= \sum_{k=-\infty}^{\infty} x[k]\, \tilde{h}_1[k-n]$$

$$d_1[m] = v[2m] = \sum_{k=-\infty}^{\infty} x[k]\, \tilde{h}_1[k-2m]$$

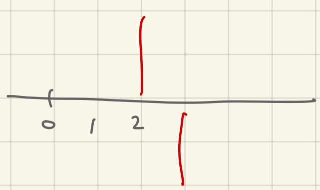
$$= \tilde{h}_1[n-2m] \cdot x[n]$$

ex) $\tilde{h}[n] = \delta[n] - \delta[n-1]$

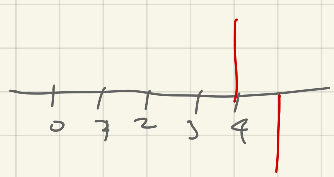
$\tilde{h}[n]$

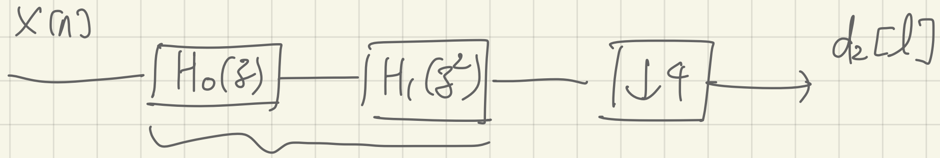


$\tilde{h}[n-2]$



$\tilde{h}[n-4]$





$$G_2(z) = H_0(z)\,H_1(z^2)$$

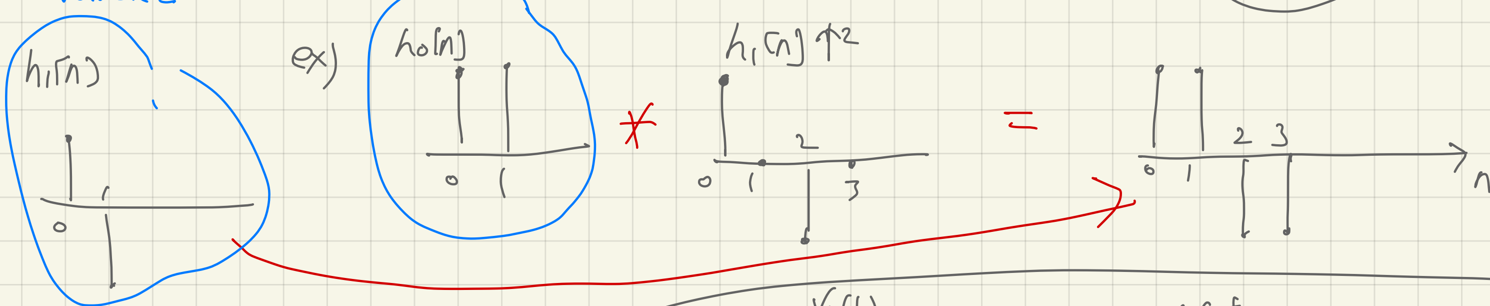
$$d_2[l] = \tilde{g}_2[n-4l] \cdot x[n]$$

mother wavelet

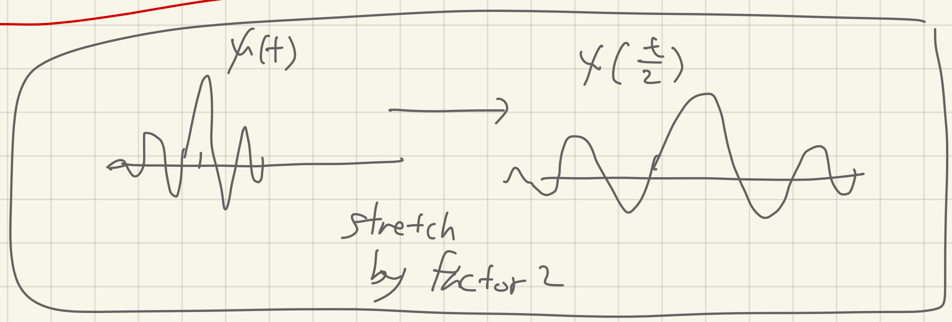
father wavelet $\tilde{g}_2[n]$

$$g_2[n] = h_0[n] * h_1[n]\uparrow 2$$

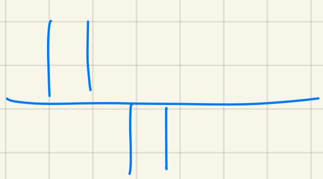
ex)



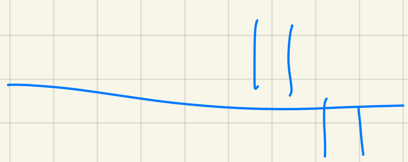
DT stretching by factor 2



$\tilde{g}[n]$



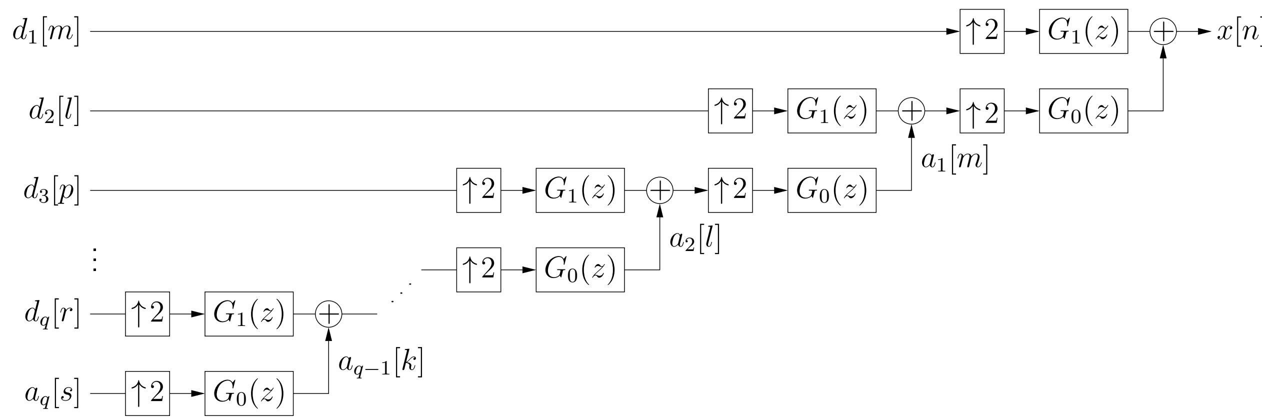
$\tilde{g}[n-4]$



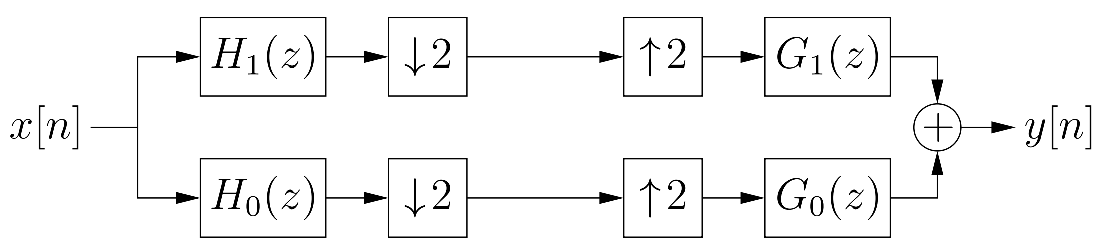
$\tilde{g}[n-8]$

The Discrete Wavelet Transform (cont.):

- For the inverse DWT (i.e., reconstruction of $x[n]$), we use an iterated synthesis filterbank



- Notice that the inverse DWT will perfectly reconstruct $x[n]$ if and only if the two-channel filterbank



  is itself perfect reconstructing, i.e., if $y[n] = x[n]$.

- We will design perfect-reconstruction (PR) filters $\{H_0(z), H_1(z), G_0(z), G_1(z)\}$ in the next note packet.

- With certain PR filters, we get an *orthogonal* DWT, i.e., $\boxed{\boldsymbol{c} = \boldsymbol{W}\boldsymbol{x}}$, where the DWT matrix $\boldsymbol{W}$ is real-valued and unitary. The inverse DWT then becomes $\boxed{\boldsymbol{x} = \boldsymbol{W}^{\mathsf{T}}\boldsymbol{c}}$.

$$d_1[m] \longrightarrow \boxed{\uparrow 2} \longrightarrow \boxed{G_1(z)} \longrightarrow y_1[n]$$

$$y_1[n] = \sum_{m=-\infty}^{\infty} d_1[m] \, g_1[n-2m]$$

$$= \begin{bmatrix} \cdots & g_1[n+2] & g_1[n] & g_1[n-2] & \cdots \end{bmatrix} \begin{bmatrix} \vdots \\ d_1[-1] \\ d_1[0] \\ d_1[1] \\ \vdots \end{bmatrix}$$